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(54) **ADJUSTABLE OPTICAL MOUNT**

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Primary Examiner — Euncha P Cherry

Related U.S. Application Data

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(51) **Int. Cl.**
G02B 26/08 (2006.01)
G02B 7/182 (2021.01)
G02B 7/198 (2021.01)

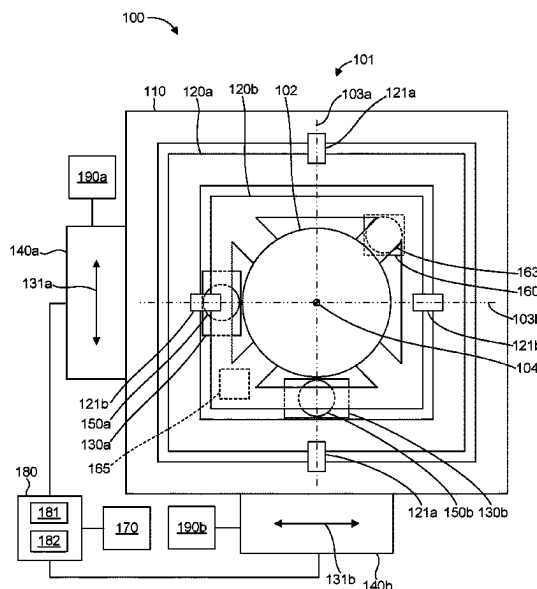
(52) **U.S. Cl.**
CPC **G02B 7/1821** (2013.01); **G02B 7/198**
(2013.01)

(58) **Field of Classification Search**
CPC G02B 7/1821; G02B 7/198; G02B 26/101;
G02B 26/08
See application file for complete search history.

(57) **ABSTRACT**

An adjustable optical mount is disclosed. The adjustable optical mount can include a base. The adjustable optical mount can also include a rotatable frame rotatably coupled to the base about an axis. The rotatable frame can be operable to support an optical element. Additionally, the adjustable optical mount can include a ramp operably coupled to the rotatable frame such that translational movement of the ramp relative to the rotatable frame causes rotation of the rotatable frame about the axis to facilitate adjustment of the optical element in a rotational degree of freedom.

26 Claims, 5 Drawing Sheets



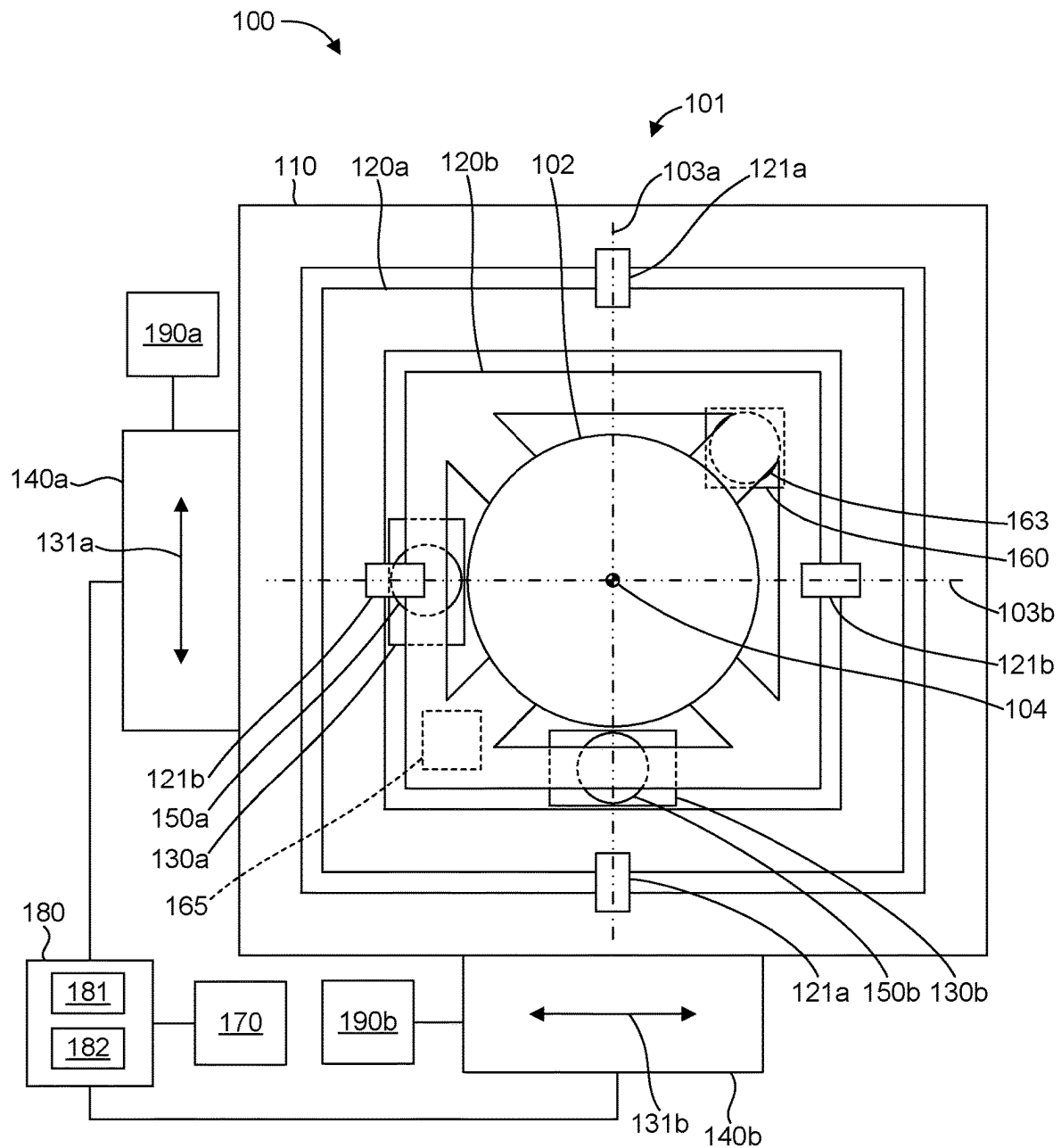


FIG. 1

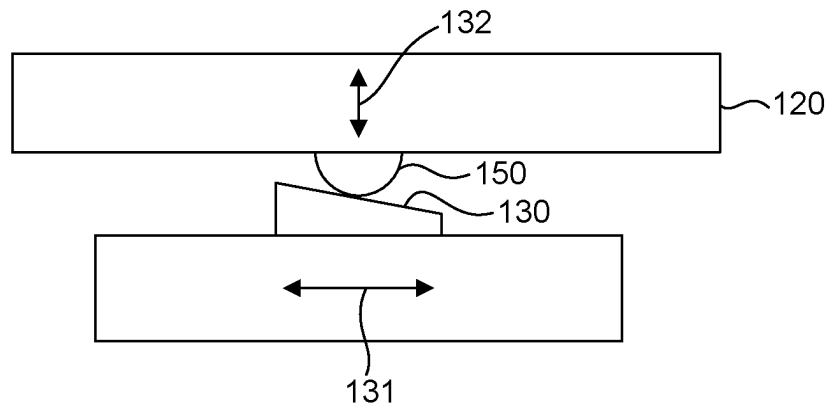


FIG. 2

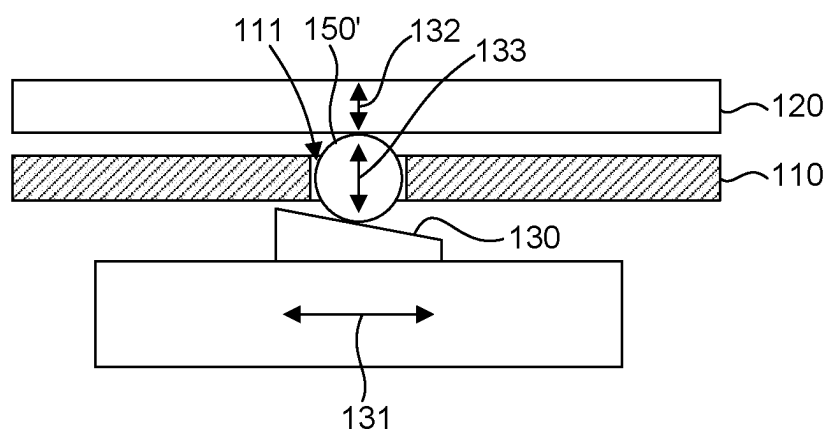


FIG. 3

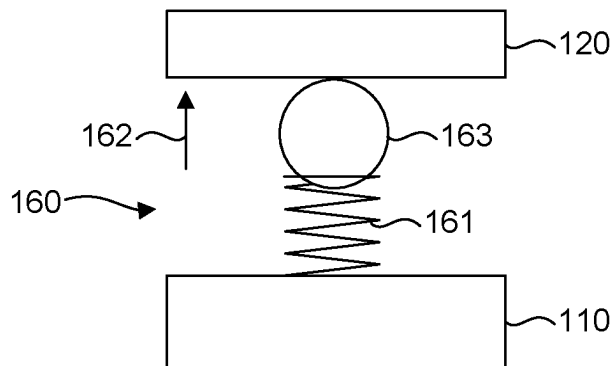


FIG. 4

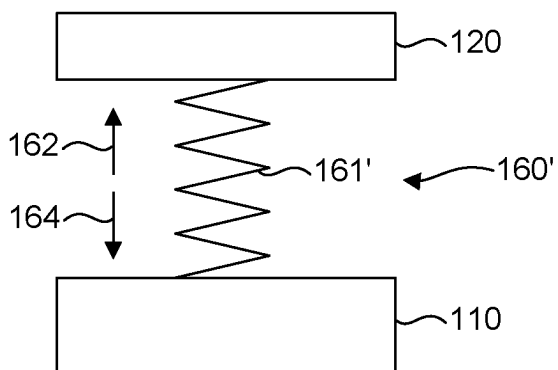


FIG. 5

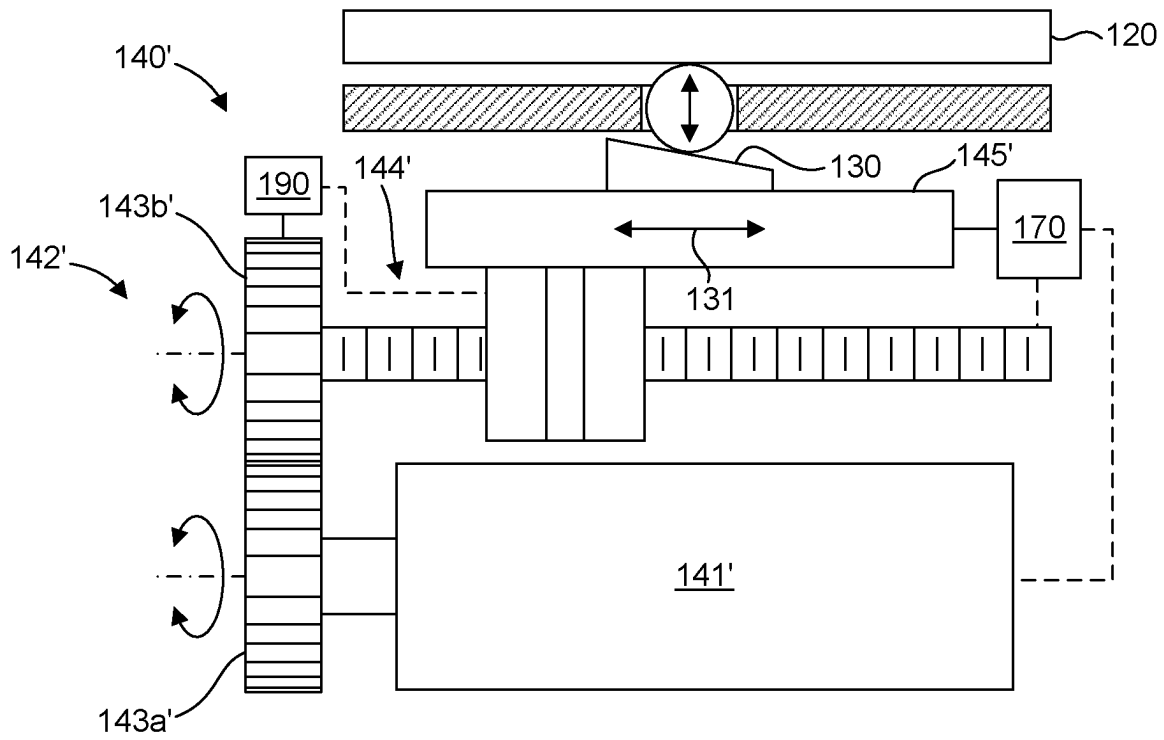


FIG. 6

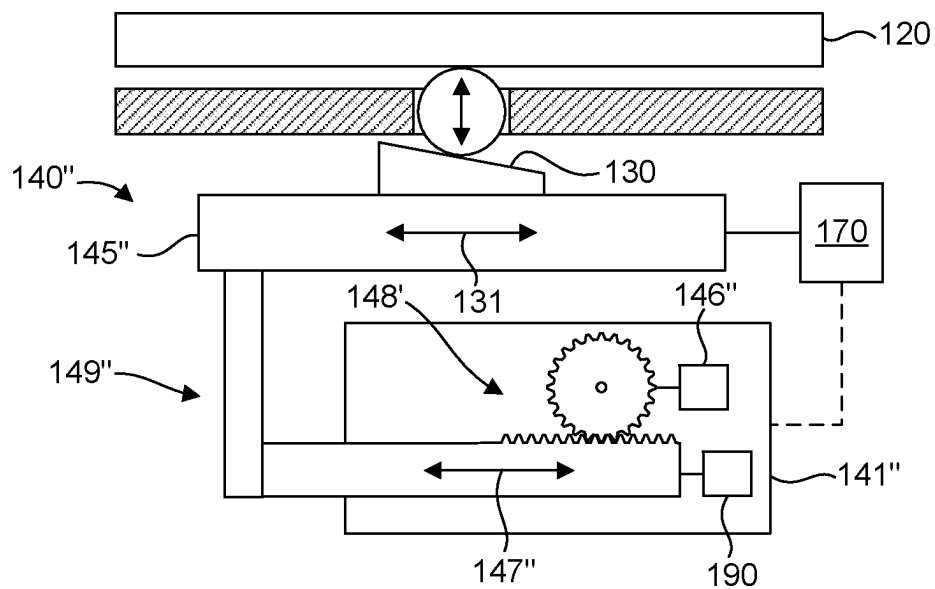


FIG. 7

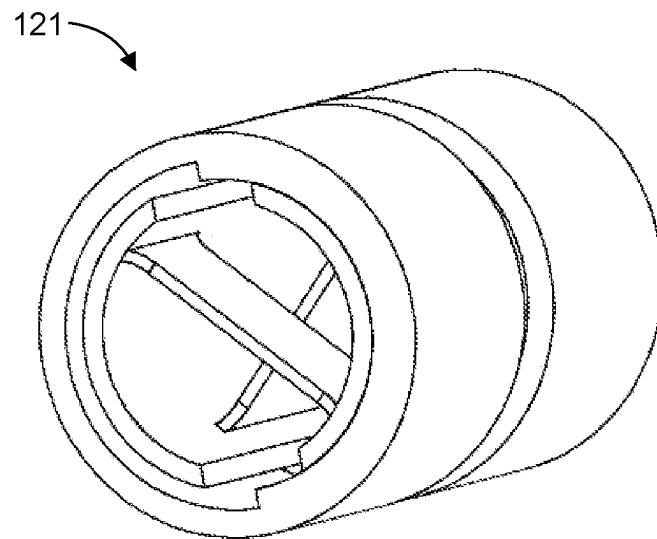


FIG. 8

ADJUSTABLE OPTICAL MOUNT

RELATED APPLICATIONS

This application claims the benefit of U.S. Provisional Patent Application No. 63/117,900, filed Nov. 24, 2020, and entitled, "Adjustable Optical Mount," which is incorporated by reference in its entirety herein.

BACKGROUND

Optical elements, such as beam splitters, mirrors, prisms, lenses, etc., are commonly used in a variety of applications. Typically, such optical elements are required to be in a precise position or alignment in order to function properly. However, the position of optical elements may slowly shift over time due to factors introduced during installation, transit, handling (e.g., drops), use (e.g., thermal effects), gimbal errors, etc. Thus, a variety of factors can lead to an optical element being misaligned both prior to and during actual field use.

BRIEF DESCRIPTION OF THE DRAWINGS

Features and advantages of the invention will be apparent from the detailed description which follows, taken in conjunction with the accompanying drawings, which together illustrate, by way of example, features of the invention; and, wherein:

FIG. 1 is a schematic top view of an adjustable optical mount system in accordance with an example of the present disclosure.

FIG. 2 is a schematic side detail view of an adjustment mechanism of the system of FIG. 1, in accordance with an example of the present disclosure.

FIG. 3 is a schematic side detail view of an adjustment mechanism of the system of FIG. 1, in accordance with an example of the present disclosure.

FIG. 4 is a schematic side detail view of a biasing mechanism of the system of FIG. 1, in accordance with an example of the present disclosure.

FIG. 5 is a schematic side detail view of a biasing mechanism of the system of FIG. 1, in accordance with an example of the present disclosure.

FIG. 6 is a schematic side detail view of an adjustment mechanism of the system of FIG. 1, in accordance with an example of the present disclosure.

FIG. 7 is a schematic side detail view of an adjustment mechanism of the system of FIG. 1, in accordance with an example of the present disclosure.

FIG. 8 illustrates a flexural pivot of the system of FIG. 1 in accordance with an example.

Reference will now be made to the exemplary embodiments illustrated, and specific language will be used herein to describe the same. It will nevertheless be understood that no limitation of the scope of the invention is thereby intended.

DETAILED DESCRIPTION

As used herein, the term "substantially" refers to the complete or nearly complete extent or degree of an action, characteristic, property, state, structure, item, or result. For example, an object that is "substantially" enclosed would mean that the object is either completely enclosed or nearly completely enclosed. The exact allowable degree of deviation from absolute completeness may in some cases depend

on the specific context. However, generally speaking the nearness of completion will be so as to have the same overall result as if absolute and total completion were obtained. The use of "substantially" is equally applicable when used in a negative connotation to refer to the complete or near complete lack of an action, characteristic, property, state, structure, item, or result.

As used herein, "adjacent" refers to the proximity of two structures or elements. Particularly, elements that are identified as being "adjacent" may be either abutting or connected. Such elements may also be near or close to each other without necessarily contacting each other. The exact degree of proximity may in some cases depend on the specific context.

An initial overview of the inventive concepts are provided below and then specific examples are described in further detail later. This initial summary is intended to aid readers in understanding the examples more quickly, but is not intended to identify key features or essential features of the examples, nor is it intended to limit the scope of the claimed subject matter.

Due to due to cleanliness or other concerns, directly accessing a misaligned optical element mounting structure may not be feasible in the field, even though it may be beneficial to correct misalignments introduced during installation, transit, handling (e.g., drops), use (e.g., thermal effects), gimbal errors, etc. Some optical systems utilize fast-steering mirrors (FSMs) or piezoelectric systems to provide high-precision adjustments of optical elements. FSMs and piezoelectric systems, however, typically require complex electronics and a constant supply of power to maintain position. Therefore, such adjustment systems are typically expensive and are not failsafe.

Accordingly, an adjustable optical mount is disclosed that, in operation, enables optical element adjustment without requiring any disassembly of the optical system. In one aspect, the adjustable optical mount can provide a low cost optical element adjustment solution that is also failsafe (e.g., does not require constant power to maintain position). The adjustable optical mount can include a base. The adjustable optical mount can also include a rotatable frame rotatably coupled to the base about an axis. The rotatable frame can be operable to support an optical element. Additionally, the adjustable optical mount can include a ramp operably coupled to the rotatable frame, such that translational movement of the ramp relative to the rotatable frame causes rotation of the rotatable frame about the axis to facilitate adjustment of the optical element in a rotational degree of freedom.

An adjustable optical mount system can comprise an optical element and an adjustable optical mount supporting the optical element. The adjustable optical mount can include a base. The adjustable optical mount can also include a rotatable frame rotatably coupled to the base about an axis. The rotatable frame can be in support of the optical element. Additionally, the adjustable optical mount can include a ramp operably coupled to the rotatable frame, such that translational movement of the ramp relative to the rotatable frame causes rotation of the rotatable frame about the axis to facilitate adjustment of the optical element in a rotational degree of freedom.

To further describe the present technology, examples are now provided with reference to the figures. With reference to FIG. 1, one embodiment of an adjustable optical mount system 100 is illustrated. The system 100 can comprise an adjustable optical mount 101 and an optical element 102 supported by the adjustable optical mount 101 for mounting

the optical element **102** to a support structure (e.g., a frame or a housing). The optical element **102** can be any suitable type of optical element or combination of optical elements, such as at least one of a beam splitter, a mirror, a prism, or a lens. In one aspect, the optical element **102** can be or include a reflective optical element (e.g., a mirror) and/or a transmissive optical element (e.g., a lens). The optical element **102** may be made of glass, metal, or another suitable material or substrate. The optical element **102** can have any suitable shape, such as a circular shape, an oval shape, a triangular shape, a rectangular shape, etc. In one aspect, the adjustable optical mount **101** can support aperture-sharing (e.g., a beam passes through the optical element **102** and a beam reflects off the optical element **102**) or fold mirror type optical elements. In another aspect, the optical element **102** can be suitable for use with any type of optical device, such as a laser, an electro-optical/infrared (EO/IR) device, etc.

The adjustable optical mount **101** can include a base **110** and a first rotatable frame **120a** rotatably coupled (at pivot joints **121a**) to the base **110** about an axis **103a**. The adjustable optical mount **101** can also include a second rotatable frame **120b** rotatably coupled (at pivot joints **121b**) to the first rotatable frame **120a** about a second axis **103b**. Thus, the second rotatable frame **120b** can be rotatably coupled to the base **110** about the axes **103a**, **103b** via rotatable couplings in the form of pivot joints **121a**, **121b** and the first rotatable frame **120a**. The second rotatable frame **120b** can interface with the optical element **102** and both the first and second rotatable frames **120a**, **120b** can support the optical element **102**. In some examples, the first rotatable frame **120a** can be associated with an elevation adjustment of the optical element **102** and the second rotatable frame **120b** can be associated with an azimuth adjustment of the optical element **102**.

The pivot joints **121a**, **121b** can be of any suitable type or configuration known in the art for rotatably coupling the rotatable frames **120a**, **120b** to one another and to the base **110**, such as at least one of a flexure, a bearing, or a bushing. Flexural pivots are devices that permit mechanical members to pivot about a common axis relative to each other through a limited angle range. Because angular motion is accomplished through flexing of elastic flexural elements, rather than contact surface displacement, flexural pivots operate without friction and thus without a need for lubrication. Flexural pivots can therefore be a substitute for bearings in applications where friction and/or the need for lubrication are concerns. A flexural pivot can utilize any suitable type of flexure element, such as a beam flexure, a wire flexure, etc. Flexural pivots typically provide angular motions of varying degrees depending upon their type. One exemplary type of a flexural pivot **121** that can be used in the adjustable optical mount **101** is shown in FIG. 8.

The adjustable optical mount **101** can further include a ramp **130a** operably coupled to the rotatable frame **120a** (indirectly as the ramp **130a** is associated with the second rotatable frame **120b** as shown in FIG. 1, but coupled to the first rotatable frame **120a** through the pivot joints **121b**), such that translational movement (e.g., in two directions **131a**) of the ramp **130a** relative to the rotatable frame **120a** causes rotation of the rotatable frame **120a** (and the rotatable frame **120b**) about the axis **103a** to facilitate adjustment of the optical element **102** in a rotational degree of freedom about the axis **103a**. The adjustable optical mount **101** can include a linear translation actuator **140a** operable to cause translational movement (e.g., in two directions **131a**) of the ramp **130a** relative to the rotatable frame **120a**.

The adjustable optical mount **101** can also include a second ramp **130b** operably coupled to the second rotatable frame **120b**, such that translational movement (e.g., in two directions **131b**) of the ramp **130b** relative to the rotatable frame **120b** causes rotation of the rotatable frame **120b** about the axis **103b** to facilitate adjustment of the optical element **102** in a rotational degree of freedom about the axis **103b**. The adjustable optical mount **101** can include a linear translation actuator **140b** operable to cause translational movement (e.g., in two directions **131b**) of the ramp **130b** relative to the rotatable frame **120b**. Thus, as the ramps **130a**, **130b** move, the rotational frames **120a**, **120b** change angular positions, which can provide precise control of the gimbaled adjustable optical mount **101** in positioning the optical element **102**.

This aspect of the present technology is further illustrated in FIGS. 2 and 3, which show a generic rotatable frame **120**, ramp **130**, and linear translation actuator **140**. As shown in FIG. 2, the ramp **130** interfaces with a spherical element or member **150**, which can be attached (e.g., affixed) to the rotatable frame **120** or integrally formed with the rotatable frame **120**. The linear translation actuator **140** can cause the ramp **130** to move in directions **131** relative to the rotatable frame **120**. As the ramp **130** moves in directions **131** while in contact with the spherical member **150**, the rotatable frame **120** moves in directions **132**. Due to a pivot joint coupling of the rotatable frame **120** (e.g., at pivot joints **121a**, **121b** in FIG. 1), the directions **132** can be rotational movement of the rotational frame **120** about an axis (e.g., the axis **103a**, **103b** in FIG. 1).

In FIG. 3, a spherical member **150'** can be a separate and distinct component that interfaces with the ramp **130** and the rotatable frame **120**. In this case, translational movement of the ramp **130** causes translational movement of the spherical member **150'** in directions **133**, which acts on the rotatable frame **120** to cause rotation of the rotatable frame **120** in the directions **132** about an axis (e.g., the axis **103a** or **103b** in FIG. 1). The spherical member **150'** can reside at least partially in a guide opening **111** of the base **110**. The guide opening **111** can be configured to constrain the spherical member **150'** to provide translational movement in the directions **133**. Thus, movement of the ramp **130** causes "up/down" movement of the spherical member **150'** within the guide opening **111** in the directions **133**, which causes the rotatable frame **120** to rotate in the directions **132** about an axis.

It should be recognized that the directions **131** of the ramp **130** can be in any suitable orientation relative to other components of the adjustable optical mount **101**. In some examples, the directions **131** can be in a plane defined by rotational axes of rotatable frames (e.g., the axes **103a**, **103b** in FIG. 1). In a particular example, the directions **131** can be parallel to the rotational axis of the rotatable frame **120**. In the example illustrated in FIG. 1, the directions **131a** can be parallel to the axis **130a**, and the directions **131b** can be parallel to the axis **130b**.

In general, and as will be recognized by those skilled in the art, the ramp **130a** and a spherical member **150a** of FIG. 1 can be associated with (i.e., caused to interface with) either one of the rotatable frames **120a**, **120b**, with the spherical member **150a** being located anywhere off the axis **103a** to provide a moment that would tend to cause rotation of the rotatable frame **120a** about the axis **103a** by action of the ramp **130a**. In the example illustrated in FIG. 1, the ramp **130a** and the spherical member **150a** are associated with the rotatable frame **120b**, but this is not intended to be limiting in any way. In this case, the spherical member **150a** is

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located on the axis **103b** so that there is no moment created by the spherical member **150a** that would tend to cause or resist rotation of the rotatable frame **120b** about the axis **103b** (e.g., by action of the ramp **130a**). Alternatively, the ramp **130a** and the spherical member **150a** can be associated with the rotatable frame **120a** to cause rotation of the rotatable frame **120a** about the axis **103a** by action of the ramp **130a**.

In general, the ramp **130b** and a spherical member **150b** of FIG. 1 can be associated with (caused to interface with) the rotatable frame **120b**, with the spherical member **150b** being located anywhere off the axis **103b** to provide a moment that would tend to cause rotation of the rotatable frame **120b** about the axis **103b** by action of the ramp **130b**. In the example illustrated in FIG. 1, the spherical member **150b** is located on the axis **103a** so that there is no moment created by the spherical member **150b** that would tend to cause or resist rotation of the rotatable frame **120a** about the axis **103a** (e.g., by action of the ramp **130b**).

In operation, or in other words as a method of positioning or adjusting an optical element associated with the adjustable optical mount **101**, one or both of the ramps **130a**, **130b** can be actuated to move (i.e., rotate) the optical element **102** about one or both of the axes **103a**, **103b**. For example, the ramp **130a** can be actuated to move in the direction **131a**, which can act on the spherical member **150a** to move the rotatable frame **120a** about the axis **103a**. The rotatable frames **120a**, **120b** are coupled to one another by the pivot joints **121b** and therefore the rotatable frames **120a**, **120b** rotate together about the axis **103a**. The optical element **102** is “carried” by the rotatable frame **120b** and is therefore also rotated about the axis **103a** by actuation of the ramp **130a**. In the illustrated example, the ramp **130a** is associated with the rotatable frame **120b**. Therefore, actuation of the ramp **130a** can cause rotation of the rotatable frame **120a** via its coupling with the rotatable frame **120b**. In other examples where the ramp **130a** is associated with the rotatable frame **120a**, actuation of the ramp **130a** can cause rotation of the rotatable frame **120b** (and the supported optical element **102**) via its coupling with the rotatable frame **120a**.

On the other hand, the ramp **130b** can be actuated to move in the direction **131b**, which can act on the spherical member **150b** to rotate the rotatable frame **120b** about the axis **103b**. The optical element **102** is “carried” by the rotatable frame **120b** and is therefore also rotated about the axis **103b** by actuation of the ramp **130b**. In this case, the rotatable frame **120b** can move independent of the rotatable frame **120a**.

In one aspect, the adjustable optical mount **101** can include one or more biasing mechanisms **160** operable to provide a biasing force on the rotatable frame **120a**, **120b** to maintain the rotatable frame **120a**, **120b** in a coupled relationship with the ramp **130a**, **130b** as the ramp **130a**, **130b** undergoes translational movement. For example, the biasing mechanism **160** can provide a biasing force to ensure that the spherical members **150a**, **150b** remain in contact with the ramps **130a**, **130b** in all operational conditions, including static and dynamic ramp **130a**, **130b** conditions. One or more biasing mechanisms **160** can be associated with one or both of the rotatable frames **120a**, **120b**. Such biasing mechanisms **160** can be located anywhere off the axis **103a**, **103b** to provide a moment that would tend to resist rotation of the rotatable frame **120a**, **120b** about the axis **103a**, **103b** by action of the ramp **130a**, **130b**. For example, a biasing mechanism **160** associated with the rotatable frame **120a** or **120b** can be located off the axis **103a** (on or off the axis **103b**) if the biasing mechanism is intended to provide resistance to rotation about the axis **103a**. Similarly, a

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biasing mechanism **160** associated with the rotatable frame **120b** can be located off the axis **103b** (on or off the axis **103a**) if the biasing mechanism is intended to provide resistance to rotation about the axis **103b**. In the example illustrated in FIG. 1, the biasing mechanism **160** is associated with the rotatable frame **120b** and is located off the axes **103a** and **103b**. Thus, a single biasing mechanism **160** can be utilized to resist rotation of the rotatable frames **120a**, **120b** about the axes **103a**, **103b**.

FIG. 4 is a detailed side view of the exemplary biasing mechanism **160** associated with a generic rotatable frame **120** and the base **110**. In general, the biasing mechanism **160** can comprise a spring **161** operably coupled to the rotatable frame **120** and the base **110** to bias the rotatable frame **120** and the base **110** away from one another (or toward one another) in the region or space proximate or adjacent to the biasing mechanism **160**. In one aspect, the spring **161** can be preloaded. In the FIG. 4 example, the spring **161** is configured to bias the rotatable frame **120** away from the base **110** in a direction **162**. In this case, the spring **161** can be a compression spring. The biasing mechanism **160** can also include a spherical member **163**, which can be biased by the spring **161**. In the illustrated example, the spring **161** is associated with (e.g., coupled or affixed to) the base **110** and the spherical member **163** is associated with (e.g., in contact with) the rotatable frame **120**, such that the spring **161** biases the spherical member **163** against the rotatable frame **120**. Alternatively, the biasing mechanism **160** can be configured such that the base **110** is associated with the rotatable frame **120** and the spherical member **163** is biased by the spring **161** against the base **110**. Thus, in one aspect, the spring **161** can be fixedly coupled to the rotatable frame **120** or the base **110**.

FIG. 5 is a detailed side view of another example of a biasing mechanism **160'** associated with the generic rotatable frame **120** and the base **110**. The biasing mechanism **160'** can include a spring **161'** operably coupled to the rotatable frame **120** and the base **110** to bias the rotatable frame **120** and the base **110** away from one another (or toward one another) in the region or space proximate the biasing mechanism **160'**. In one aspect, the spring **161'** can be preloaded. In the FIG. 5 example, the spring **161'** can be configured to bias the rotatable frame **120** away from the base **110** in the direction **162**. In this case, the spring **161'** can be a compression spring. Alternatively, the spring **161'** can be configured to bias the rotatable frame **120** toward the base **110** in a direction **164**. In this case, the spring **161'** can be a tension spring. A tension spring configuration can be utilized at a location where the rotatable frame **120** and the base **110** are to be pulled toward one another in order to maintain contact between the spherical members **150a**, **150b** and the ramps **130a**, **130b**, such as at **165** in FIG. 1. In the FIG. 5 example, the spring **161'** is associated with the base **110** and the rotatable frame **120**, such that the spring **161'** can be coupled to or merely in contact with either component. Thus, in one aspect, the spring **161'** can be fixedly coupled to the rotatable frame **120**, the base **110**, or both.

It should be recognized that any suitable biasing mechanism can be utilized in accordance with the principles disclosed herein. For example, as an alternative to those examples discussed above, a biasing mechanism can be incorporated into or with one or more pivot joints **121a**, **121b**, such as by incorporating a spring with the joint or preloading a flexural pivot.

In one aspect, the adjustable optical mount **101** can be a kinematic mount in that the mount **101** can support the optical element **102** in six degrees of freedom (translations

along three mutually orthogonal axes, and three rotations around these axes) without over-constraining the optical element **102** (over-constraining can result in distortion and uncertain positioning of the optical element **102**). For example, each ramp **130a**, **130b** can be associated with a point of contact for constraining the optical element **102** (e.g., via the spherical members **150a**, **150b**) and the biasing mechanism **160** can provide a third point of contact for constraining the optical element **102**. The ramp **130a** can be operable to move the rotatable frame **120a** about the axis **103a**, the ramp **130b** can be operable to move the rotatable frame **120b** about the axis **103b**, and the biasing mechanism **160** can provide resistance to movement of both rotatable frames **120a**, **120b** about axes **103a**, **103b** to ensure that no looseness or “play” exists in the movement of the rotatable frames **120a**, **120b** about the axes **103a**, **103b**. In one aspect of a kinematic mount, as illustrated in FIG. 1, a center of gravity **104** of the optical element **102** can be centrally located between the contact points associated with the ramps **130a**, **130b** and the biasing mechanism **160**.

In one aspect, illustrated in FIG. 2, the spherical member **150** can provide for a single point of contact with the rotatable frame **120**. In this example, the ramp **130** can interface directly with the rotatable frame **120** (of which the spherical member **150** is an integral part or affixed thereto) to cause rotation of the rotatable frame **120** about an axis (e.g., the axis **103a** or **103b**). FIG. 3 illustrates another example of single point contact with the rotatable frame **120**. In this example, the spherical member **150'** can be a separate and distinct component that interfaces with the ramp **130** and the rotatable frame **120**. In some examples, a spherical member as disclosed herein can be a ball bearing (e.g., commercially available ball bearings), which may be inexpensive, made to tight tolerances, and substantially burr-free. Interfacing or contact surfaces of the ramp **130** and rotatable frame **120** with the spherical member **150'** can have any suitable configuration, such as a flat or planar configuration, to provide stable, predictable movement of the spherical member **150'** along such surfaces during operation. For the same reasons, an interfacing or contact surface of the rotatable frame **120** (or the base **110**) with the spherical member **163** of the biasing mechanism **160** can be similarly configured.

Referring again to FIG. 1, in some examples, the system **100** can include a sensor **170** operable to facilitate determining a position (e.g., angular position) of the optical element **102**. The sensor **170** can be or include any suitable sensor type or configuration known in the art that can be used to determine a position of the optical element **102**. For example, the sensor **170** can comprise at least one of a potentiometer, a position sensor, a distance sensor, an encoder, a Hall effect sensor, or an eddy current sensor. In some examples, the sensor **170** can be operable to determine a position of the optical element **102** directly (e.g., a laser beam reflecting off the optical sensor **102**) and/or indirectly (e.g., via determining a position of the rotatable frame **120a**, **120b**, the ramp **130a**, **130b**, the linear translation actuator **140a**, **140b**, or other suitable feature or component of the adjustable optical mount **101**). The sensor **170** can be operable to provide active feedback on the position of the optical element **102**.

In some examples, the system **100** can include a control system **180** operably coupled to the linear translation actuators **140a**, **140b** and the sensor **170** to facilitate control of the linear translation actuators **140a**, **140b** based on the position of the optical element **102** as determined based on sensor **170** input or information. In one aspect, this configuration

can enable active feedback control of the linear translation actuators **140a**, **140b** to ensure that the optical element **102** is correctly aligned or positioned. The control system **180** can include any suitable hardware (e.g., a processor **181** and memory **182**), software, etc. typically used in control systems to enable open loop and/or closed loop control of the linear translation actuators **140a**, **140b**. In one aspect, “slow” misalignment effects, such as gimbal orbital error, can be calibrated out via a lookup table.

In some examples, the adjustable optical mount **101** can include anti-backlash mechanisms **190a**, **190b** operable to minimize backlash in movement of respective ramps **130a**, **130b**. Aspects of the anti-backlash mechanism are discussed in more detail below.

The linear translation actuators **140a**, **140b** can be or include any suitable actuator type or configuration known in the art that can be used to provide translational (e.g., linear) movement of the ramps **130a**, **130b**, respectively. For example, the linear translation actuators **140a**, **140b** can comprise at least one of a linear actuator (e.g., a linear motor), a rotary actuator, a lead screw, a ball screw, a worm and worm gear, or an electric motor (e.g., a servomotor, a stepper motor, a piezo drive, etc.).

FIG. 6 illustrates the ramp **130** and rotatable frame **120** configuration of FIG. 3, with a more particular linear translation actuator **140'** configuration. In this case, the linear translation actuator **140'** can include a rotary actuator **141'** or other suitable rotary output device, such as a rotary electric motor. The linear translation actuator **140'** can also include a drive train **142'** operably coupled to the rotary actuator **141'**. The drive train **142'** can include gears **143a'**, **143b'** coupled to a suitable rotational to linear motion conversion device **144'** as known in the art, such as a leadscrew, a ball screw, a rack and pinion gear, etc. The linear translation actuator **140'** can also include a linear slide **145'** coupled to the rotational to linear motion conversion device **144'**. The linear slide **145'** can be constrained to move linearly in the directions **131** and can be coupled to the ramp **130**. The linear slide **145'** can be of any suitable design or configuration as known in the art, which may include a carriage, a rail, a track, a bearing, a bushing, etc. to provide smooth and consistent linear movement of the ramp **130** in the directions **131**, depending upon the direction of rotation of the rotary actuator **141'**.

In one aspect, one or more sensors **170** (e.g., an encoder, potentiometer, etc.) can be operably coupled to or otherwise associated with at least one component of the linear translation actuator **140'**. In particular, one or more sensors **170** can be operably coupled to or otherwise associated with at least one of the rotary actuator **141'**, a component of the drive train **142'** (e.g., a gear **143a'**, **143b'**, a leadscrew or ball screw **144'**, etc.), or the linear slide **145'**. For example, one or more sensors **170** can be configured to follow the movement of a coupled or associated component (e.g., via gearing) to determine a position or change of position of the coupled or associated component, which can facilitate determining a position (e.g., angular position) of the optical element **102**.

In one aspect, one or more anti-backlash mechanisms or systems **190** can be operably coupled to or otherwise associated with at least one component of the linear translation actuator **140'**. In particular, one or more anti-backlash mechanisms **190** can be operably coupled to or otherwise associated with at least one of the rotary actuator **141'**, a component of the drive train **142'** (e.g., a gear **143a'**, **143b'**, a leadscrew or ball screw **144'**, etc.), or the linear slide **145'**. For example, one or more anti-backlash mechanisms **190**

can be configured to minimize or eliminate backlash in any drive train interface (e.g., a gear or screw interface). Any suitable anti-backlash mechanism known in the art may be utilized, such as an anti-backlash ball screw, an anti-backlash lead screw, anti-backlash gears, etc. Such anti-backlash mechanisms typically utilize a biasing spring or a "back torque" applied by a motor operable to bias a gear or screw component against an interfacing gear or screw component.

FIG. 7 illustrates the ramp **130** and rotatable frame **120** configuration of FIG. 3, with another more particular linear translation actuator **140** configuration. In this case, the linear translation actuator **140** can include a linear actuator **141** or other suitable linear output device, such as a linear electric motor, operable to provide an output in two linear directions **147**. In some examples, such as in the example illustrated in FIG. 7, the linear actuator **141** can include a rotary electric motor **146** coupled to a rack and pinion gear train **148**, which can be configured to move in the linear directions **147**.

The linear translation actuator **140** can also include a drive arm **149** operably coupled to linear actuator **141**. The linear translation actuator **140** can also include a linear slide **145** coupled to the drive arm **149**. The linear slide **145** can be constrained to move linearly in the directions **131** and can be coupled to the ramp **130**. The linear slide **145** can be of any suitable design or configuration as known in the art, which may include a carriage, a rail, a track, a bearing, a bushing, etc. to provide smooth and consistent linear movement of the ramp **130** in the directions **131**.

In one aspect, one or more sensors **170** (e.g., an encoder, potentiometer, etc.) can be operably coupled to or otherwise associated with at least one component of the linear translation actuator **140**. In particular, one or more sensors **170** can be operably coupled to or otherwise associated with at least one of the linear actuator **141** (e.g., the rotary electric motor **146** and/or the rack and pinion gear train **148**), the drive arm **149**, or the linear slide **145**.

In one aspect, one or more anti-backlash mechanisms **190** can be operably coupled to or otherwise associated with at least one component of the linear translation actuator **140**. In particular, one or more anti-backlash mechanisms **190** can be operably coupled to or otherwise associated with at least one of the linear actuator **141** (e.g., the rotary electric motor **146** and/or the rack and pinion gear train **148**), the drive arm **149**, or the linear slide **145**. For example, one or more anti-backlash mechanisms **190** can be configured to minimize or eliminate backlash in any drive train interface (e.g., an interface of the rack and pinion gear train **148**).

In one aspect, the linear translation actuators **140a**, **140b** of FIG. 1 can maintain the respective ramps **130a**, **130b** in a fixed position even when the linear translation actuators **140a**, **140b** are not powered. For example, the linear translation actuators **140a**, **140b** can include components or features that prevent or resist movement when not powered. For instance, certain mechanical components, such as a lead screw, a ball screw, a worm and worm gear, etc., can have mechanical advantage and/or friction that resists or prevents movement under forces applied to the linear translation actuator **140a**, **140b** via one or more of the ramps **130a**, **130b** (e.g., from gimbaled components such as the rotatable frames **120a**, **120b**) under static or dynamic conditions. Thus, power is not required to maintain position of the optical element **102** and the linear translation actuators **140a**, **140b** can provide a failsafe in the event that power is lost. Even if the optical element **102** does shift position, this can be detected by the sensor(s) **170** and corrected by operation of the linear translation actuators **140a**, **140b** (e.g.,

as controlled by the control system **180**). The technology disclosed herein can therefore provide a simple way to adjust the optical element **102** at any time (e.g., in the field) without requiring disassembly of the system **100** (e.g., breaking seals, etc.). This attribute is beneficial for many applications, such as high precision applications and cryogenic applications.

In one aspect, the adjustable optical mount **101** as disclosed herein can utilize many commercial off-the-shelf (COTS) parts (e.g., ball bearings, actuator components (e.g., motors, gears, ball screws, lead screws, linear slides, etc.), sensors, etc.), which can provide low complexity in the design of the adjustable optical mount **101** and still provide highly precise and repeatable adjustments. If improved performance is desired, more precise manufacturing techniques can be utilized (e.g., lapping interface surfaces) and/or more precise sensors can be included.

There is also provided, in some examples, various methods for facilitating the adjustment of an optical element by configuring any of the adjustable optical mounts and adjustable optical mount systems (including the optical element) described herein. In accordance with one example, a method for facilitating adjustment of an optical element is disclosed. The method can comprise rotatably coupling a frame to a base about an axis. Optionally, the method can further comprise rotatably coupling a second rotatable frame about a second axis. The first and second rotatable frames can be rotatably coupled (at pivot joints) to one another about the axes, and relative to the base. In one aspect, the second rotatable frame can be rotatably coupled to the base about the axes via the rotatable couplings (e.g., in the form of pivot joints) and the first rotatable frame. The second rotatable frame can be configured to interface with the optical element. The method can further facilitate the first rotatable frame being associated with an elevation adjustment of the optical element, and the second rotatable frame being associated with an azimuth adjustment of the optical element.

The method can also comprise facilitating support of the optical element with one or both of the first and second rotatable frames. The method can further comprise operably coupling (indirectly) a first ramp to the first rotatable frame such that translational movement of the first ramp relative to the first rotatable frame causes rotation of the first rotatable frame (and the second rotatable frame if present) about the first axis. Additionally, the method can comprise facilitating translational movement of the first ramp relative to the first rotatable frame to rotate the first rotatable frame (and the second rotatable frame if present) about the first axis and facilitate adjustment of the optical element in a first rotational degree of freedom. The method can optionally further comprise operably coupling a second ramp to the second rotatable frame such that translational movement of the second ramp relative to the second rotatable frame causes rotation of the second rotatable frame about the second axis. Additionally, the method can comprise facilitating translational movement of the second ramp relative to the second rotatable frame to rotate the second rotatable frame about the second axis and facilitate adjustment of the optical element in a second rotational degree of freedom. It is noted that no specific order is required in this method, though generally in one embodiment, these method steps can be carried out sequentially.

In one aspect of the method, facilitating movement of the ramp(s) can comprise operably coupling a linear translation actuator to the ramp (or ramps, respectively). In another aspect of the method, operably coupling the ramp(s) to the rotatable frame can comprise interfacing a spherical member

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with the ramp and the rotatable frame (or with the ramps and the rotatable frames, respectively), and constraining the spherical member to provide translational movement.

The method can further comprise configuring the adjustable optical mount, and particularly a linear translation actuator, with an anti-backlash mechanism as described herein for minimizing backlash in movement of an associated ramp.

The method can further comprise configuring the adjustable optical mount with a biasing mechanism operable to provide a biasing force on the rotatable frame(s) to maintain the rotatable frame(s) in a coupled relationship with the ramp as the ramp undergoes translational movement.

Reference was made to the examples illustrated in the drawings and specific language was used herein to describe the same. It will nevertheless be understood that no limitation of the scope of the technology is thereby intended. Alterations and further modifications of the features illustrated herein and additional applications of the examples as illustrated herein are to be considered within the scope of the description.

Although the disclosure may not expressly disclose that some embodiments or features described herein may be combined with other embodiments or features described herein, this disclosure should be read to describe any such combinations that would be practicable by one of ordinary skill in the art. The user of "or" in this disclosure should be understood to mean non-exclusive or, i.e., "and/or," unless otherwise indicated herein.

Furthermore, the described features, structures, or characteristics may be combined in any suitable manner in one or more examples. In the preceding description, numerous specific details were provided, such as examples of various configurations to provide a thorough understanding of examples of the described technology. It will be recognized, however, that the technology may be practiced without one or more of the specific details, or with other methods, components, devices, etc. In other instances, well-known structures or operations are not shown or described in detail to avoid obscuring aspects of the technology.

Although the subject matter has been described in language specific to structural features and/or operations, it is to be understood that the subject matter defined in the appended claims is not necessarily limited to the specific features and operations described above. Rather, the specific features and acts described above are disclosed as example forms of implementing the claims. Numerous modifications and alternative arrangements may be devised without departing from the spirit and scope of the described technology.

What is claimed is:

1. An adjustable optical mount, comprising:

a base;

a first rotatable frame rotatably coupled to the base about an axis, the first rotatable frame configured to support an optical element;

a ramp operably coupled to the first rotatable frame such that translational movement of the ramp relative to the first rotatable frame causes rotation of the first rotatable frame about the axis to facilitate adjustment of the optical element in a rotational degree of freedom; and

a biasing mechanism configured to provide a biasing force on the first rotatable frame to maintain the first rotatable frame in a coupled relationship with the ramp as the ramp undergoes the translational movement.

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2. The adjustable optical mount of claim 1, wherein the ramp interfaces directly with the first rotatable frame to cause rotation of the first rotatable frame about the axis.

3. The adjustable optical mount of claim 1, further comprising a spherical member interfacing with the ramp and the first rotatable frame, wherein the translational movement of the ramp causes translational movement of the spherical member that acts on the first rotatable frame to cause rotation of the first rotatable frame about the axis.

4. The adjustable optical mount of claim 3, wherein the spherical member resides at least partially in an opening of the base, the opening configured to constrain the spherical member to provide the translational movement of the spherical member.

5. The adjustable optical mount of claim 1, further comprising a linear translation actuator configured to provide the translational movement of the ramp relative to the first rotatable frame.

6. The adjustable optical mount of claim 5, wherein the linear translation actuator comprises at least one of a linear actuator, a rotary actuator, a lead screw, a ball screw, or an electric motor.

7. The adjustable optical mount of claim 5, further comprising an anti-backlash mechanism configured to minimize backlash in movement of the ramp.

8. The adjustable optical mount of claim 1, further comprising a sensor configured to facilitate determining a position of the optical element.

9. The adjustable optical mount of claim 8, wherein the sensor comprises at least one of a potentiometer, a position sensor, a distance sensor, an encoder, a Hall effect sensor, or an eddy current sensor.

10. The adjustable optical mount of claim 1, wherein the biasing mechanism comprises a spring.

11. The adjustable optical mount of claim 10, wherein the biasing mechanism further comprises a spherical member biased by the spring against the first rotatable frame.

12. The adjustable optical mount of claim 10, wherein the spring is fixedly coupled to the first rotatable frame.

13. The adjustable optical mount of claim 1, wherein the first rotatable frame is rotatably coupled to the base via at least one of a flexure, a bearing, or a bushing.

14. The adjustable optical mount of claim 1, further comprising:

a second rotatable frame rotatably coupled to the first rotatable frame about a second axis; and

a second ramp operably coupled to the second rotatable frame such that translational movement of the second ramp relative to the second rotatable frame causes rotation of the second rotatable frame about the second axis to facilitate adjustment of the optical element in a second rotational degree of freedom.

15. The adjustable optical mount of claim 14, wherein the first rotatable frame or the second rotatable frame interfaces with the optical element.

16. The adjustable optical mount of claim 14, further comprising a spherical member interfacing with the second ramp and the second rotatable frame, wherein the translational movement of the second ramp causes translational movement of the spherical member that acts on the second rotatable frame to cause rotation of the second rotatable frame about the second axis.

17. An adjustable optical mount system, comprising:
an optical element; and

an adjustable optical mount supporting the optical element, the adjustable optical mount including:
a base,

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- a first rotatable frame rotatably coupled to the base about a first axis, the first rotatable frame in support of the optical element,
- a first ramp operably coupled to the first rotatable frame such that translational movement of the first ramp relative to the first rotatable frame causes rotation of the first rotatable frame about the first axis to facilitate adjustment of the optical element in a rotational degree of freedom,
- a second rotatable frame rotatably coupled to the first rotatable frame about a second axis, and
- a second ramp operably coupled to the second rotatable frame such that translational movement of the second ramp relative to the second rotatable frame causes rotation of the second rotatable frame about the second axis to facilitate adjustment of the optical element in a second rotational degree of freedom.
18. The system of claim 17, further comprising a spherical member interfacing with the first ramp and the first rotatable frame, wherein the translational movement of the first ramp causes translational movement of the spherical member that acts on the first rotatable frame to cause rotation of the first rotatable frame about the first axis.
19. The system of claim 18, wherein the spherical member resides at least partially in an opening of the base, the opening configured to constrain the spherical member to provide the translational movement of the spherical member.
20. The system of claim 17, wherein the first rotatable frame or the second rotatable frame interfaces with the optical element.
21. The system of claim 17, further comprising:
- a linear translation actuator configured to provide the translational movement of the first ramp relative to the first rotatable frame;
 - a sensor configured to facilitate determining a position of the optical element; and

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- a control system operably coupled to the linear translation actuator and the sensor to facilitate control of the linear translation actuator based on the position of the optical element.
22. A method for facilitating adjustment of an optical element, comprising:
- rotatably coupling a frame to a base about an axis;
 - supporting the optical element with the frame;
 - operably coupling a ramp to the frame such that translational movement of the ramp relative to the frame causes rotation of the frame about the axis, wherein operably coupling the ramp to the frame comprises interfacing a spherical member with the ramp and the frame and constraining the spherical member to provide translational movement; and
 - facilitating the translational movement of the ramp relative to the frame to rotate the frame about the axis and facilitate adjustment of the optical element in a rotational degree of freedom.
23. The method of claim 22, wherein facilitating the translational movement of the ramp comprises operably coupling a linear translation actuator to the ramp.
24. The method of claim 22, wherein the translational movement of the ramp causes the translational movement of the spherical member, which acts on the frame to cause rotation of the frame about the axis.
25. The system of claim 17, further comprising a biasing mechanism operable to provide a biasing force on the first rotatable frame to maintain the first rotatable frame in a coupled relationship with the first ramp as the first ramp undergoes translational movement.
26. The method of claim 22, further comprising providing a biasing force on the frame to maintain the frame in a coupled relationship with the ramp as the ramp undergoes translational movement.

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